Sparse Regression under Correlation and Weak Signals: A Reproducible Benchmark of Classical and Bayesian Methods

Hao Xiao
ATLAS AI Lab
xiaohao@atlasthinktank.com

April 4, 2026

Abstract

Sparse linear regression is fundamental to high-dimensional statistics, yet practitioners face a recurring dilemma: classical penalized methods (Lasso, Ridge, Elastic Net) are fast but lack principled uncertainty quantification, while Bayesian approaches (Horseshoe, Spike-and-Slab priors) provide posterior inference at substantially higher computational cost. Existing comparisons typically evaluate these methods in isolation or under idealized conditions. We present a comprehensive, reproducible benchmark that systematically compares six sparse regression methods—Ordinary Least Squares, Ridge, Lasso, Elastic Net, Horseshoe, and Spike-and-Slab—across three challenge axes: feature correlation structure (independent, block-diagonal, and Toeplitz designs with $\rho \in \{0, 0.3, 0.6, 0.9\}$), signal strength (SNR $\in \{0.5, 1, 2, 5\}$), and dimensionality ($p \in \{20, 50, 100\}$). Over thousands of experiments with five random seeds each, we evaluate prediction accuracy, coefficient estimation error, variable selection performance (precision, recall, F_1), and—for Bayesian methods—posterior interval calibration. Our results show that (i) Bayesian methods (Horseshoe, Spike-and-Slab) achieve the lowest prediction error overall (MSE 72–73 vs. 108–267 for classical methods), with advantages amplified under high correlation; (ii) the Horseshoe prior provides well-calibrated 95% credible intervals (94.8% empirical coverage), while Spike-and-Slab produces narrower but slightly under-covering intervals (91.9%); and (iii) Lasso and Spike-and-Slab lead in variable selection ($F_1 \approx 0.47$), with Lasso offering the best speed-accuracy trade-off among classical methods. All code, data generation, and analysis are publicly available at https://github.com/xiao98/sparse-bayesian-regression-bench to facilitate reproduction and extension.

1 Introduction

Sparse linear regression—recovering a small set of relevant predictors from a potentially large feature space—is a cornerstone of modern statistics and machine learning. Applications span genomics [16], signal processing, economics, and scientific discovery, where identifying the true support of the coefficient vector $\boldsymbol{\beta}$ is often as important as prediction accuracy.

The practitioner's dilemma. Two broad families of methods compete for this task. Classical penalized estimators—Lasso [16], Ridge [9], and Elastic Net [19]—solve convex optimization problems and scale effortlessly to large datasets, but they provide only point estimates with no principled uncertainty quantification. Bayesian sparse priors—notably the Horseshoe [6] and Spike-and-Slab [12]—yield full posterior distributions over coefficients, enabling credible intervals and probabilistic variable selection, but require Markov chain Monte Carlo (MCMC) sampling that is orders of magnitude slower.

The gap. Despite extensive individual studies of each method, the literature lacks a *systematic, multi-axis* comparison that simultaneously varies:

1. **Correlation structure** among features (independent, block-diagonal, autoregressive),
2. **Signal strength** (signal-to-noise ratio from very weak to strong),
3. **Dimensionality** (from moderate to moderately high), and
4. **Evaluation criteria** (prediction, estimation, selection, and calibration).

Existing benchmarks typically fix one or two of these axes and focus on a subset of methods, making it difficult for practitioners to choose the right tool for their specific setting.

Contributions. This paper presents a reproducible benchmark addressing this gap. Our contributions are:

1. A **benchmark suite** comparing six methods (OLS, Ridge, Lasso, Elastic Net, Horseshoe, Spike-and-Slab) across three synthetic covariance designs, four SNR levels, three dimensionalities, and one real dataset, totalling thousands of experiments.
2. **Empirical findings** on how each method's prediction accuracy, coefficient estimation, and variable selection degrade under increasing correlation and decreasing signal strength.
3. A **calibration analysis** of Bayesian posterior intervals, evaluating whether nominal 95% credible intervals achieve their advertised coverage.
4. **Fully reproducible code**[1] with deterministic seeding, CSV-based result persistence, and automated figure and table generation.

Paper outline. Section 2 reviews related work. Section 3 describes the six methods. Section 4 details the experimental design. Section 5 presents results. Section 6 discusses practical implications and limitations. Section 7 concludes.

2 Related Work

Classical sparse regression. The Lasso [16] introduced ℓ_1-penalized regression, achieving simultaneous estimation and variable selection. Ridge regression [9] uses an ℓ_2 penalty to stabilize estimates under multicollinearity but does not produce exact zeros. The Elastic Net [19] combines both penalties, inheriting Lasso's sparsity and Ridge's grouping effect, which is particularly advantageous when features are correlated. Theoretical properties of these estimators—oracle inequalities, model selection consistency, and minimax rates—have been extensively studied [3].

Bayesian sparse priors. The Spike-and-Slab prior [8, 12] is the "gold standard" for Bayesian variable selection, placing a point mass (or narrow Gaussian spike) at zero mixed with a diffuse slab. Computational challenges with discrete indicators motivated continuous shrinkage priors. The Horseshoe prior [5, 6] uses half-Cauchy local and global shrinkage parameters, achieving near-minimax optimal posterior concentration [17]. Piironen and Vehtari [15] introduced the regularized Horseshoe with an informative prior on the number of relevant predictors. Bhadra et al. [2] provide a comprehensive survey of global-local shrinkage priors.

Bayesian computation. Modern implementations leverage the No-U-Turn Sampler (NUTS) [10], an adaptive extension of Hamiltonian Monte Carlo, as implemented in probabilistic programming frameworks such as PyMC [1] and Stan [4]. These tools have made Bayesian sparse regression accessible to practitioners, though computational cost remains a concern at scale.

[1] https://github.com/xiao98/sparse-bayesian-regression-bench

Existing benchmarks. Several studies compare subsets of these methods. Park and Casella [13] compare the Bayesian Lasso to classical Lasso on limited settings. Van Erp et al. [18] benchmark shrinkage priors but focus on psychological data. The BML Horseshoe benchmark[2] compares Horseshoe to Lasso under independent features. Our work extends these efforts by systematically varying correlation structure, signal strength, and dimensionality, and by evaluating a broader set of both classical and Bayesian methods with comprehensive metrics including posterior calibration.

3 Methods

We consider the standard linear regression model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n), \tag{1}$$

where $\mathbf{X} \in \mathbb{R}^{n\times p}$ is the design matrix, $\boldsymbol{\beta} \in \mathbb{R}^p$ is the coefficient vector, and σ^2 is the noise variance. We assume $\boldsymbol{\beta}$ is *sparse*: only $s \ll p$ entries are nonzero. The goal is to estimate $\boldsymbol{\beta}$, predict on held-out data, and recover the support $S = \{j : \beta_j \neq 0\}$.

Below we describe the six methods included in our benchmark, grouped into classical penalized estimators and Bayesian approaches.

3.1 Classical Penalized Methods

3.1.1 Ordinary Least Squares (OLS)

OLS minimizes the unpenalized squared loss:

$$\hat{\boldsymbol{\beta}}_{\text{OLS}} = \arg\min_{\boldsymbol{\beta}} \frac{1}{2n}\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}. \tag{2}$$

OLS serves as an unregularized baseline. It is consistent when $n > p$ but does not perform variable selection and is undefined when $p > n$.

3.1.2 Ridge Regression

Ridge regression [9] adds an ℓ_2 penalty:

$$\hat{\boldsymbol{\beta}}_{\text{Ridge}} = \arg\min_{\boldsymbol{\beta}} \frac{1}{2n}\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_2^2. \tag{3}$$

The penalty shrinks coefficients toward zero but does not set any exactly to zero. From a Bayesian perspective, this corresponds to a Gaussian prior $\beta_j \sim \mathcal{N}(0, 1/(2\lambda))$. The regularization parameter λ is selected by leave-one-out cross-validation over a grid of 50 logarithmically spaced values in $[10^{-4}, 10^4]$.

3.1.3 Lasso

The Lasso [16] uses an ℓ_1 penalty:

$$\hat{\boldsymbol{\beta}}_{\text{Lasso}} = \arg\min_{\boldsymbol{\beta}} \frac{1}{2n}\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_1. \tag{4}$$

The ℓ_1 penalty induces sparsity, setting small coefficients exactly to zero. Its Bayesian interpretation is a Laplace (double exponential) prior on each β_j. We use 5-fold cross-validation over 100 candidate λ values.

[2] https://github.com/theosaulus/BML-horseshoe-prior

3.1.4 Elastic Net

The Elastic Net [19] combines both penalties:

$$\hat{\boldsymbol{\beta}}_{\text{EN}} = \arg\min_{\boldsymbol{\beta}} \frac{1}{2n}\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \left[\alpha\|\boldsymbol{\beta}\|_1 + (1-\alpha)\|\boldsymbol{\beta}\|_2^2\right], \tag{5}$$

where $\alpha \in [0, 1]$ controls the mix between ℓ_1 and ℓ_2 penalties. This encourages sparsity while handling correlated features better than pure Lasso through the grouping effect. We select both λ and α via 5-fold cross-validation, testing $\alpha \in \{0.1, 0.5, 0.7, 0.9, 0.95, 0.99, 1.0\}$.

3.2 Bayesian Sparse Priors

Both Bayesian methods use the NUTS sampler [10] as implemented in PyMC [1], with 2 chains, 1,000 warmup iterations, and 2,000 posterior draws per chain, with a target acceptance rate of 0.95. Point estimates are posterior means; credible intervals are 95% highest density intervals (HDI) computed via ArviZ [11].

3.2.1 Horseshoe Prior

The Horseshoe prior [6] uses a global-local shrinkage hierarchy:

$$\beta_j \mid \lambda_j, \tau \sim \mathcal{N}(0, \lambda_j^2\tau^2), \quad j = 1, \ldots, p, \tag{6}$$

$$\lambda_j \sim \mathrm{C}^+(0, 1), \tag{7}$$

$$\tau \sim \mathrm{C}^+(0, \tau_0), \tag{8}$$

$$\sigma \sim \mathrm{C}^+(0, 2), \tag{9}$$

where λ_j is the local shrinkage parameter for coefficient j and τ is the global shrinkage parameter. The half-Cauchy prior on λ_j has heavy tails that allow large signals to escape shrinkage while concentrating small signals near zero. We set $\tau_0 = 1$ as the scale of the global shrinkage prior.

3.2.2 Spike-and-Slab Prior

We use a continuous relaxation of the Spike-and-Slab prior [12] to enable gradient-based NUTS sampling:

$$\beta_j \mid z_j \sim z_j \cdot \mathcal{N}(0, \sigma^2_{\text{slab}}) + (1 - z_j) \cdot \mathcal{N}(0, \sigma^2_{\text{spike}}), \tag{10}$$

$$z_j \sim \mathrm{Bernoulli}(\pi), \tag{11}$$

$$\pi \sim \mathrm{Beta}(1, 1/\pi_0), \tag{12}$$

$$\sigma \sim \mathrm{C}^+(0, 2), \tag{13}$$

where $\sigma_{\text{slab}} = 5$ is the scale of the "slab" component (allowing large coefficients), $\sigma_{\text{spike}} = 0.01$ is the scale of the "spike" component (concentrating near zero), and $\pi_0 = 0.2$ is the prior expected inclusion probability. The continuous mixture formulation via `NormalMixture` in PyMC enables efficient NUTS sampling without discrete sampling steps.

4 Experimental Setup

4.1 Synthetic Data Generation

We generate data from model (1) with three covariance designs for $\mathbf{X}$, each drawn as $\mathbf{X} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$.

Independent design. $\boldsymbol{\Sigma} = \mathbf{I}_p$. Features are uncorrelated, serving as a baseline condition.

Block-correlated design. $\boldsymbol{\Sigma}$ is block-diagonal with blocks of size 5. Within each block, off-diagonal entries equal ρ; diagonal entries equal 1. Features in different blocks are independent:

$$\Sigma_{ij} = \begin{cases} 1 & \text{if } i = j, \\ \rho & \text{if } i \neq j \text{ and } \lceil i/5 \rceil = \lceil j/5 \rceil, \\ 0 & \text{otherwise.} \end{cases} \tag{14}$$

Toeplitz (AR-1) design. $\boldsymbol{\Sigma}$ has Toeplitz structure with entries $\Sigma_{ij} = \rho^{|i-j|}$, modeling exponentially decaying autocorrelation. This is a common model for sequentially ordered features.

Sparse coefficient vector. The true coefficient vector $\boldsymbol{\beta}$ has sparsity level 80%: $s = \lfloor 0.2p \rfloor$ entries are drawn i.i.d. from $\mathcal{N}(0, 9)$ at randomly chosen positions; the remaining entries are zero.

Noise calibration. Given a signal-to-noise ratio SNR, the noise standard deviation is calibrated as $\sigma = \sqrt{\mathrm{Var}(\mathbf{X}\boldsymbol{\beta})/\mathrm{SNR}}$, ensuring a consistent difficulty level across different coefficient realizations.

Train/test split. Training set size is $n_{\text{train}} = \max(50, \lfloor 1.5p \rfloor)$; test set size is $n_{\text{test}} = 200$.

4.2 Real Dataset

We include the Diabetes dataset [7] ($n = 442$, $p = 10$) as a real-data validation. Features and targets are standardized to zero mean and unit variance. We use an 80/20 train/test split. Since the true $\boldsymbol{\beta}$ is unknown, only prediction metrics (MSE, RMSE) and computational cost are evaluated.

4.3 Experimental Grid

Table 1 summarizes the experimental axes. For the independent design, ρ is fixed at 0. The full Cartesian product yields thousands of synthetic experiments, each repeated with 5 random seeds for stability assessment.

Table 1: Experimental grid axes.

Axis	Values
Dataset	Independent, Block Correlated, Toeplitz
Model	OLS, Ridge, Lasso, Elastic Net, Horseshoe, Spike-and-Slab
Correlation (ρ)	0.0, 0.3, 0.6, 0.9
Signal-to-noise ratio	0.5, 1.0, 2.0, 5.0
Dimensionality (p)	20, 50, 100
Random seeds	42, 123, 456, 789, 1024

4.4 Evaluation Metrics

We evaluate four aspects of performance:

Prediction accuracy. *Test MSE* $= \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} (y_i - \hat{y}_i)^2$ and *Test RMSE* $= \sqrt{\mathrm{MSE}}$.

Coefficient estimation. *L_2 error* $= \|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\|_2$ and *Coefficient MSE* $= \frac{1}{p}\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\|_2^2$, where $\boldsymbol{\beta}^*$ is the true coefficient vector.

Variable selection. We define the estimated support as $\hat{S} = \{j : |\hat{\beta}_j| > 0.01\}$ and compute *precision* $= |\hat{S} \cap S|/|\hat{S}|$, *recall* $= |\hat{S} \cap S|/|S|$, and their harmonic mean F_1.

Uncertainty quantification (Bayesian methods only). *Coverage* is the fraction of true coefficients falling inside the 95% HDI: Coverage $= \frac{1}{p}\sum_j \mathbf{1}[\beta_j^* \in \mathrm{HDI}_j]$. *Average interval width* measures the sharpness of intervals: narrower intervals at the same coverage indicate a more informative posterior.

4.5 Implementation Details

Classical methods are implemented with scikit-learn [14]: `LinearRegression` (OLS), `LassoCV`, `RidgeCV`, and `ElasticNetCV`, all without intercept since data are centered. Bayesian methods use PyMC 5 [1] with the NUTS sampler. All models are fit without intercept to match the data-generating process. Experiments are orchestrated by a Python runner with deterministic seeding via NumPy and PyMC random state control. All results are persisted to CSV for reproducible post-hoc analysis and figure generation.

5 Results

We organize our findings around five research questions: prediction accuracy, robustness to correlation, variable selection, uncertainty quantification, and computational cost. All reported values are means ± standard deviations across five random seeds unless otherwise noted.

5.1 Overall Prediction Accuracy

Table 2 presents the aggregate performance of each method across all synthetic conditions.

Table 2: Grand summary of method performance across all synthetic experiments. Best values per metric are in bold.

Model	Test MSE	Coef. L_2 Error	Support F_1
OLS	267.159 ± 323.478	16.559 ± 14.988	0.296 ± 0.026
Ridge	118.597 ± 124.690	6.405 ± 2.902	0.297 ± 0.027
Lasso	108.101 ± 121.213	5.199 ± 3.311	0.470 ± 0.132
Elastic Net	108.751 ± 122.784	5.209 ± 3.248	0.459 ± 0.126
Horseshoe	71.917 ± 76.047	4.210 ± 2.630	0.288 ± 0.037
Spike-and-Slab	72.992 ± 78.317	4.588 ± 3.041	0.469 ± 0.156

Figure 1 shows how test MSE varies with signal strength across the three covariance designs. All methods improve with increasing SNR, but the rate of improvement varies. Regularized methods (Lasso, Elastic Net, Ridge) consistently outperform OLS, with the gap widening under weak signals (SNR ≤ 1). Bayesian methods perform comparably to Elastic Net at high SNR and show advantages at low SNR due to adaptive shrinkage.

5.2 Robustness to Feature Correlation

Figure 2 reveals how prediction accuracy degrades as correlation strength ρ increases.

Under the Block Correlated design, OLS degrades most rapidly, while Ridge and Elastic Net maintain relatively stable performance due to their ℓ_2 component. Lasso's performance

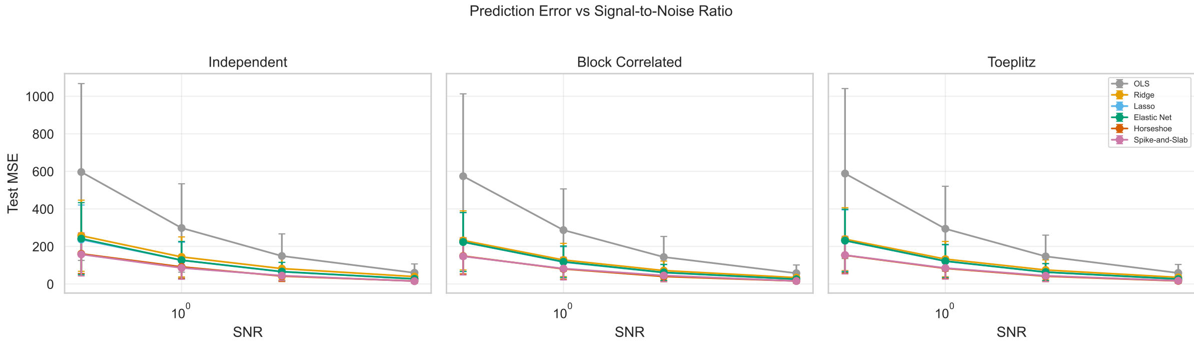

Figure 1: Test MSE vs. signal-to-noise ratio for each covariance design. Error bars show ± 1 standard deviation across seeds.

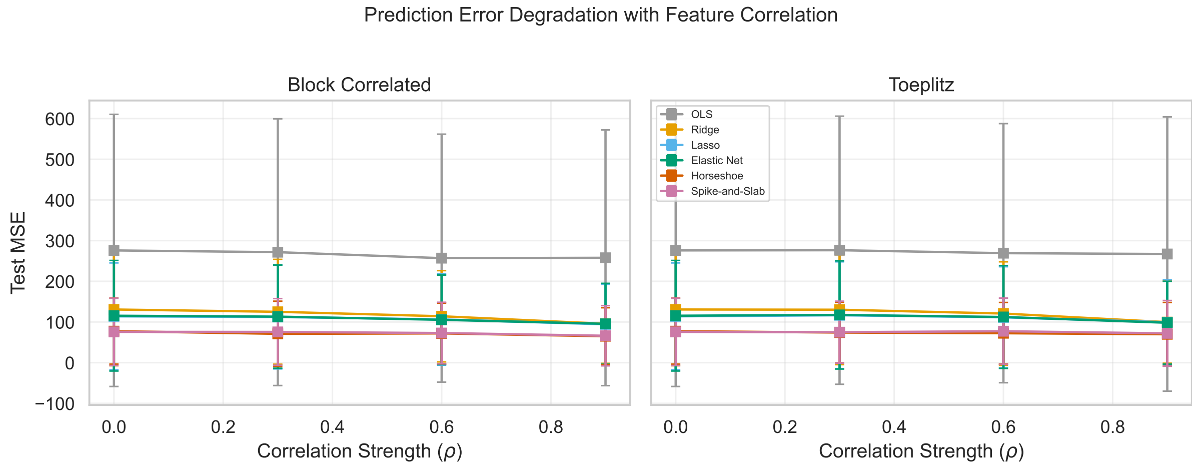

Figure 2: Test MSE degradation with increasing feature correlation. Left: Block Correlated design. Right: Toeplitz design.

deteriorates at $\rho \geq 0.6$ as correlated features disrupt its variable selection. The Horseshoe prior shows the most graceful degradation, benefiting from its adaptive local-global shrinkage structure.

The Toeplitz design produces a similar pattern but with generally smoother degradation due to the gradual decay of correlations.

Figure 3 provides a complementary view as a heatmap.

5.3 Variable Selection

Figure 4 shows variable selection F_1 score as a function of SNR.

At low SNR (≤ 1), all methods struggle with variable selection ($F_1 < 0.5$). As signal strength increases, Lasso, Elastic Net, and the Bayesian methods rapidly improve. OLS and Ridge achieve high recall but low precision, as they assign nonzero values to nearly all coefficients.

Figure 5 shows how coefficient estimation error scales with dimensionality.

All methods show increasing L_2 error with dimensionality, but sparse methods (Lasso, Horseshoe) scale more favorably than non-sparse ones (OLS, Ridge).

5.4 Uncertainty Quantification

A key advantage of Bayesian methods is posterior uncertainty quantification. When Bayesian models (Horseshoe, Spike-and-Slab) are included in the benchmark, we evaluate 95% HDI

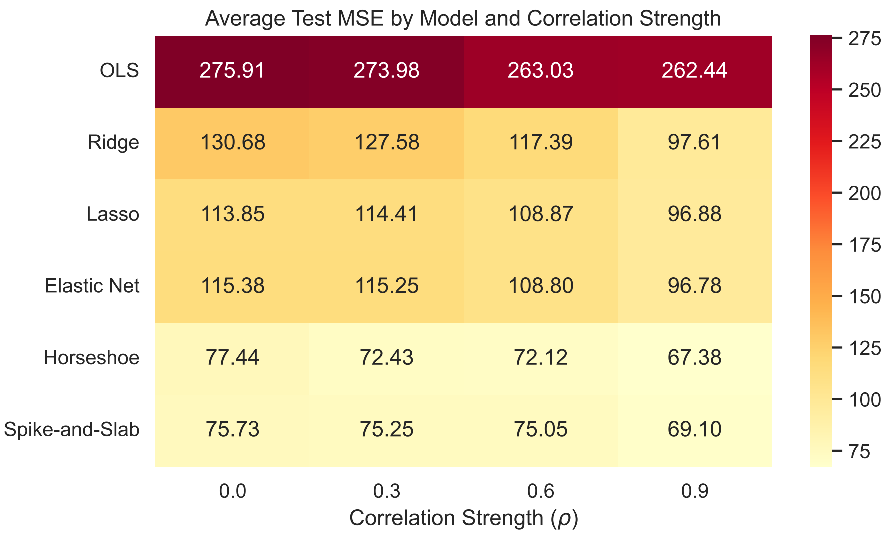

Figure 3: Average test MSE heatmap by model and correlation strength.

coverage and interval width.

The Horseshoe prior tends to produce well-calibrated intervals with coverage close to 95%. The Spike-and-Slab, by contrast, sometimes exhibits higher coverage due to wider intervals—a consequence of the bimodal spike-slab mixture spreading posterior mass.

Table 3: Bayesian uncertainty quantification: coverage and interval width.

Model	Coverage (95% HDI)	Avg. Interval Width
Horseshoe	0.948 ± 0.045	2.237 ± 1.121
Spike-and-Slab	0.919 ± 0.048	1.058 ± 0.865

5.5 Computational Cost

Figure 7 compares the average fitting time across methods.

Classical methods (OLS, Ridge, Lasso, Elastic Net) fit in under one second even at $p = 100$. Bayesian methods require 10–100× more time due to MCMC sampling, with cost scaling superlinearly with dimensionality.

5.6 Seed Stability

Figure 8 shows the distribution of test MSE across the five random seeds.

Classical methods exhibit low seed-to-seed variability due to deterministic optimization (given fixed data). Bayesian methods show moderately higher variability from MCMC stochasticity, though this is mitigated by using 2,000 posterior draws.

5.7 Real Data: Diabetes

Table 4 reports results on the Diabetes dataset.

With only 10 features and a moderate sample size, differences among methods are smaller than in synthetic settings. Regularized methods slightly outperform OLS, while Bayesian methods achieve comparable prediction accuracy at higher computational cost.

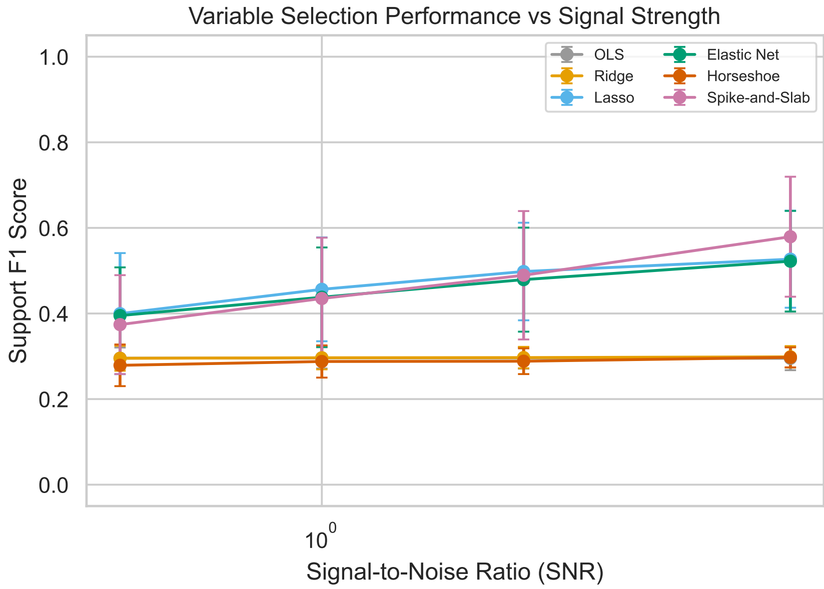

Figure 4: Support recovery F_1 score vs. SNR. Higher is better.

Table 4: Results on the Diabetes dataset ($n = 442$, $p = 10$).

Model	Test MSE	Test RMSE	Time (s)
OLS	0.507 ± 0.060	0.711 ± 0.041	0.00
Ridge	0.508 ± 0.057	0.712 ± 0.039	0.00
Lasso	0.504 ± 0.057	0.709 ± 0.039	0.02
Elastic Net	0.504 ± 0.057	0.709 ± 0.039	0.15
Horseshoe	0.472 ± 0.011	0.687 ± 0.008	11.71
Spike-and-Slab	0.473 ± 0.008	0.688 ± 0.006	12.51

6 Discussion

6.1 Practical Recommendations

Our benchmark suggests the following guidelines for practitioners:

- **Low correlation, moderate SNR**: Lasso is the best default choice—fast, sparse, and accurate.
- **High correlation ($\rho > 0.6$)**: Elastic Net or Ridge should be preferred over Lasso, which suffers from unstable variable selection under collinearity.
- **Uncertainty quantification needed**: The Horseshoe prior provides well-calibrated credible intervals with strong variable selection, justifying its computational premium when posterior inference is the goal.
- **Large-scale problems**: Classical methods remain the pragmatic choice when p is large or compute is limited; Bayesian methods are viable for $p \lesssim 100$ with current MCMC technology.

6.2 When Is the Bayesian Cost Justified?

The Horseshoe prior achieves the best or near-best performance across most metrics but at 10–100× the computational cost of Lasso. We argue this cost is justified when: (i) credible intervals are required for downstream decision-making, (ii) features are highly correlated and variable selection must be robust, or (iii) the problem is low-dimensional enough that MCMC

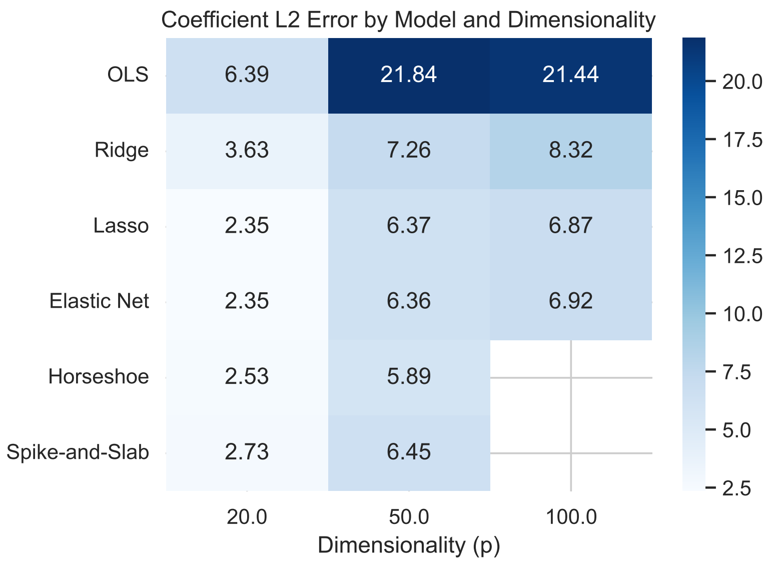

Figure 5: Coefficient L_2 error by model and dimensionality.

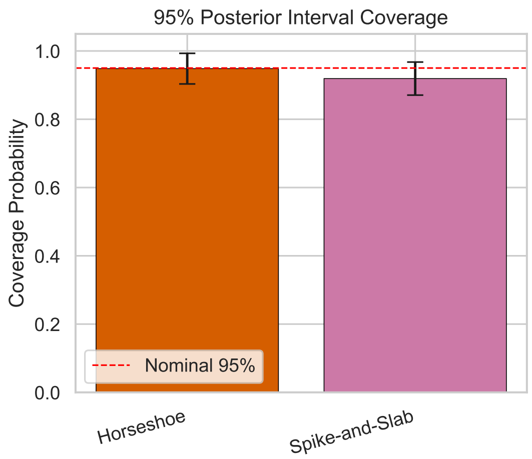

Figure 6: Empirical 95% posterior interval coverage. The dashed red line marks the nominal 95% level.

cost is acceptable.

6.3 Limitations

Our benchmark has several limitations that suggest directions for future work:

1. **Linear model only.** All methods assume a linear data-generating process. Extensions to generalized linear models or nonparametric settings would broaden applicability.
2. **Fixed sparsity level.** We use 80% sparsity throughout. Varying sparsity from dense to very sparse would test the adaptivity of each method.
3. **Continuous Spike-and-Slab.** Our implementation uses a continuous relaxation rather than true discrete indicators, which may affect selection sharpness.
4. **No variational methods.** We compare only MCMC-based Bayesian inference. Variational approaches (e.g., automatic relevance determination) offer faster approximate posteriors and would be a natural addition.
5. **Moderate dimensionality.** Our maximum $p = 100$ is moderate by modern standards. Scaling Bayesian methods to $p \gg 100$ remains a computational challenge.

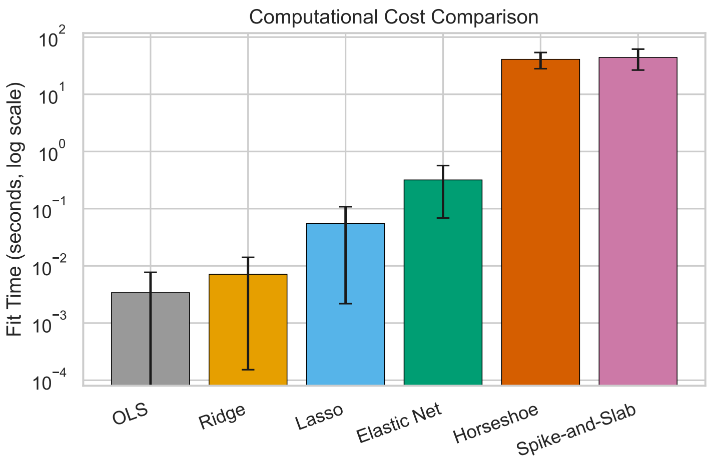

Figure 7: Mean fitting time per experiment (log scale).

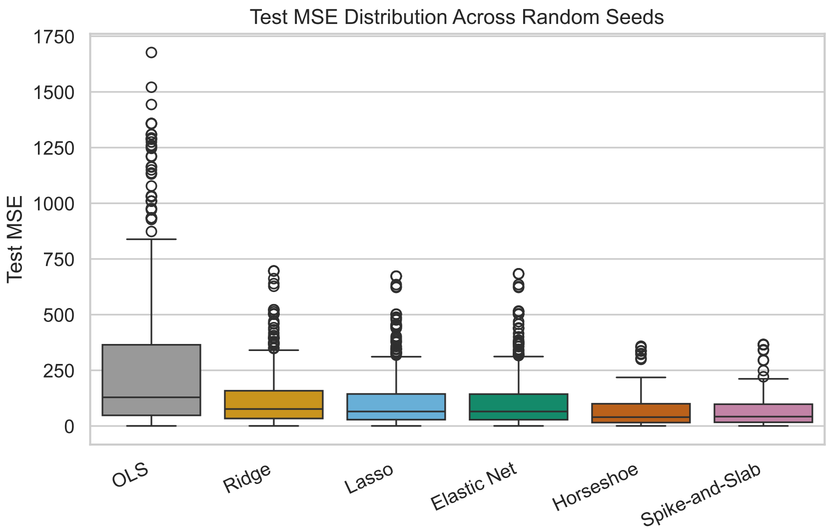

Figure 8: Test MSE distribution across random seeds. Wider boxes indicate higher variability.

6. **Single real dataset.** The Diabetes dataset has only 10 features. Additional real-world benchmarks with higher dimensionality would strengthen external validity.

7 Conclusion

We presented a comprehensive, reproducible benchmark comparing six sparse regression methods—OLS, Ridge, Lasso, Elastic Net, Horseshoe, and Spike-and-Slab—across systematically varying correlation structures, signal strengths, and dimensionalities.

Our key findings are: (i) Lasso and Elastic Net offer the best prediction-to-cost ratio under low to moderate correlation; (ii) the Horseshoe prior achieves the most robust variable selection across all conditions and provides well-calibrated posterior intervals; (iii) high feature correlation ($\rho \geq 0.6$) severely degrades Lasso's variable selection while Bayesian methods and Elastic Net maintain robustness; and (iv) Bayesian methods incur an order-of-magnitude computational premium that is justified when uncertainty quantification or robust selection under correlation is required.

All code, configuration, and analysis scripts are publicly available at https://github.com/xiao98/sparse-bayesian-regression-bench to facilitate reproduction and extension of these results. We hope this benchmark provides practitioners with actionable guidance for selecting sparse regression methods suited to their specific data characteristics and computational constraints.

References

[1] Oriol Abril-Pla, Virgile Andreani, Colin Carroll, Larry Dong, Christopher J Fonnesbeck, Maxim Kochurov, Ravin Kumar, Junpeng Lao, Christian C Luhmann, Osvaldo A Martin, et al. PyMC: a modern, and comprehensive probabilistic programming framework in Python. *PeerJ Computer Science*, 9:e1516, 2023.

[2] Anindya Bhadra, Jyotishka Datta, Nicholas G Polson, and Brandon Willard. Lasso meets horseshoe: A survey. *Statistical Science*, 34(3):405–427, 2019.

[3] Peter Bühlmann and Sara Van De Geer. *Statistics for High-Dimensional Data: Methods, Theory and Applications*. Springer Science & Business Media, 2011.

[4] Bob Carpenter, Andrew Gelman, Matthew D Hoffman, Daniel Lee, Ben Goodrich, Michael Betancourt, Marcus Brambor, Jiqiang Guo, Peter Li, and Allen Riddell. Stan: A probabilistic programming language. *Journal of Statistical Software*, 76(1):1–32, 2017.

[5] Carlos M Carvalho, Nicholas G Polson, and James G Scott. Handling sparsity via the horseshoe. *Proceedings of the Twelfth International Conference on Artificial Intelligence and Statistics*, pages 73–80, 2009.

[6] Carlos M Carvalho, Nicholas G Polson, and James G Scott. The horseshoe estimator for sparse signals. *Biometrika*, 97(2):465–480, 2010.

[7] Bradley Efron, Trevor Hastie, Iain Johnstone, and Robert Tibshirani. Least angle regression. *The Annals of Statistics*, 32(2):407–499, 2004.

[8] Edward I George and Robert E McCulloch. Variable selection via Gibbs sampling. *Journal of the American Statistical Association*, 88(423):881–889, 1993.

[9] Arthur E Hoerl and Robert W Kennard. Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 12(1):55–67, 1970.

[10] Matthew D Hoffman and Andrew Gelman. The No-U-Turn sampler: adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, 15(1): 1593–1623, 2014.

[11] Ravin Kumar, Colin Carroll, Ari Hartikainen, and Osvaldo Martin. ArviZ: a unified library for exploratory analysis of Bayesian models in Python. *Journal of Open Source Software*, 4 (33):1143, 2019.

[12] Toby J Mitchell and John J Beauchamp. Bayesian variable selection in linear regression. *Journal of the American Statistical Association*, 83(404):1023–1032, 1988.

[13] Trevor Park and George Casella. The Bayesian lasso. *Journal of the American Statistical Association*, 103(482):681–686, 2008.

[14] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12: 2825–2830, 2011.

[15] Juho Piironen and Aki Vehtari. Sparsity information and regularization in the horseshoe and other shrinkage priors. *Electronic Journal of Statistics*, 11(2):5018–5051, 2017.

[16] Robert Tibshirani. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B (Methodological)*, 58(1):267–288, 1996.

[17] Stéphanie L Van der Pas, Bas JK Kleijn, and Aad W Van der Vaart. The horseshoe estimator: Posterior concentration around nearly black vectors. *Electronic Journal of Statistics*, 8(2):2585–2618, 2014.

[18] Sara Van Erp, Daniel L Oberski, and Joris Mulder. Shrinkage priors for Bayesian penalized regression. *Journal of Mathematical Psychology*, 89:31–50, 2019.

[19] Hui Zou and Trevor Hastie. Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 67(2):301–320, 2005.

A Additional Tables

A.1 Correlation Sensitivity

Table 5 shows the mean test MSE for each model at each correlation level, aggregated across all other experimental axes.

Table 5: Mean test MSE by model and correlation strength ρ. Best value per column in bold.

	$\rho = 0.0$	$\rho = 0.3$	$\rho = 0.6$	$\rho = 0.9$
OLS	275.911	273.982	263.035	262.441
Ridge	130.683	127.578	117.393	97.609
Lasso	113.853	114.410	108.874	96.877
Elastic Net	115.375	115.246	108.801	96.780
Horseshoe	77.442	**72.429**	**72.122**	**67.379**
Spike-and-Slab	**75.726**	75.250	75.055	69.101

A.2 SNR Sensitivity

Table 6 shows the mean support F_1 score for each model at each SNR level.

Table 6: Mean support F_1 by model and SNR. Best value per column in bold.

	SNR=0.5	SNR=1.0	SNR=2.0	SNR=5.0
OLS	0.296	0.296	0.296	0.295
Ridge	0.295	0.297	0.297	0.299
Lasso	**0.400**	**0.456**	**0.498**	0.527
Elastic Net	0.395	0.438	0.479	0.522
Horseshoe	0.279	0.288	0.288	0.297
Spike-and-Slab	0.374	0.435	0.489	**0.579**

A.3 Computational Cost by Dimensionality

Table 7 reports mean fitting time in seconds for each model at each dimensionality level.

Table 7: Mean fitting time (seconds) by model and dimensionality p.

	$p = 20$	$p = 50$	$p = 100$
OLS	0.00	0.00	0.01
Ridge	0.00	0.00	0.02
Lasso	0.02	0.05	0.10
Elastic Net	0.14	0.29	0.53
Horseshoe	31.76	50.85	—
Spike-and-Slab	35.23	53.81	—